Experiment 4: The Sol-Gel Technique – Lab Report

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1) Aim

To synthesize SiO_2 nanopowder by the sol-gel method, coat it by dip and spin coating, and examine the effect of SiO_2 content on the contact angle and viscosity of PVA solutions.

2) Materials and Devices

TEOS, ethanol, distilled water, NH_4F, NH_4OH, PVA, Triton X-100; magnetic stirrer, balance, drying oven, furnace, laser, spin coater, FTO/ITO glass, Ostwald viscometer.

3) Experimental Procedure

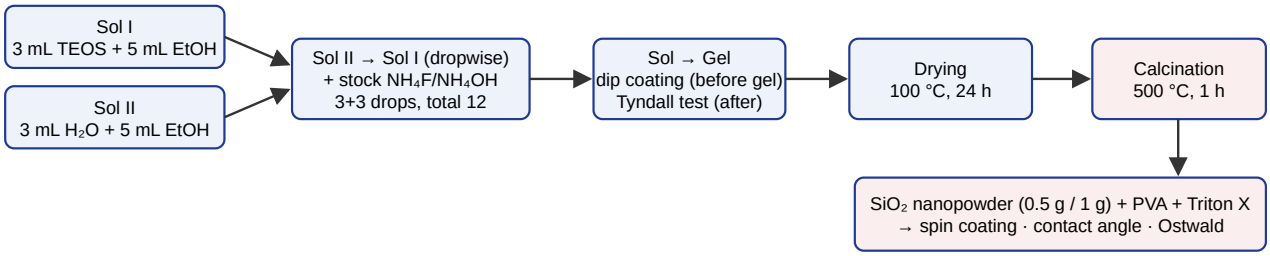


Figure 1. Flow chart of the experiment.

- Solution II (3 mL water + 5 mL ethanol) was added dropwise to Solution I (3 mL TEOS + 5 mL ethanol) under stirring; hydrolysis started (≡Si–OR + H_2O → ≡Si–OH + ROH).
- Stock catalyst (1.9 g NH_4F + 23 mL NH_4OH in 100 mL water [1]) was added 3 drops at a time until a coating formed on a dipped glass slide (12 drops in total).
- Before gelation a glass slide was dipped into the sol and slowly withdrawn (dip coating). After gelation a laser beam was passed through a beaker of water and then through the gel.
- The wet gel was weighed, dried at 100 °C for 24 h, calcined at 500 °C for 1 h and weighed after each step to obtain the SiO_2 nanopowder.
- PVA suspensions with 0.5 g and 1 g SiO_2 were spin-coated (3 drops, 500/800 rpm, 5/10 s). One drop of water was placed on each coating, and the viscosities were measured with the Ostwald viscometer (5 mL, water as reference).

4) Results and Discussion

4.1) Percent weight loss

Step	Mass (g)	Loss (%)
Wet gel	10.900	–
Dried (100 °C, 24 h)	0.576	94.72
Calcined (500 °C, 1 h)	0.394	31.60
Total		**96.39**

$$Loss\ (\%) = (m_{before} - m_{after}) / m_{before} \times 100$$

Drying: (10.9 − 0.576)/10.9 × 100 = 94.72 %
Calcination: (0.576 − 0.394)/0.576 × 100 = 31.60 %
Total: (10.9 − 0.394)/10.9 × 100 = 96.39 %

Most of the mass was lost during drying, because the wet gel is mostly liquid (water, ethanol and the ethanol released by hydrolysis) trapped in the silica network. During calcination, residual organic groups (–OC_2H_5) burned off and Si–OH groups condensed into Si–O–Si, releasing water. Theoretically, 3 mL TEOS (0.0134 mol) gives about 0.81 g SiO_2, so the yield was ≈ 49 %; the rest was used for dip coating or remained in the beaker and crucible.

4.2) Tyndall effect

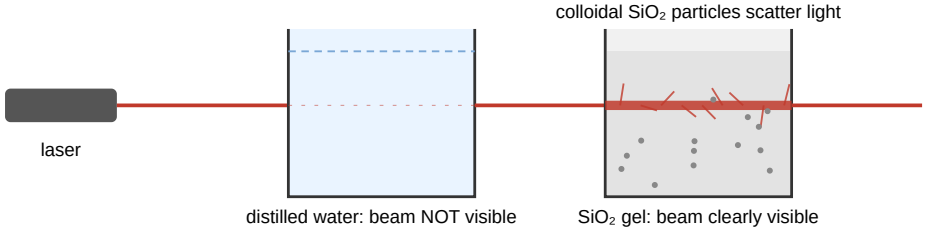
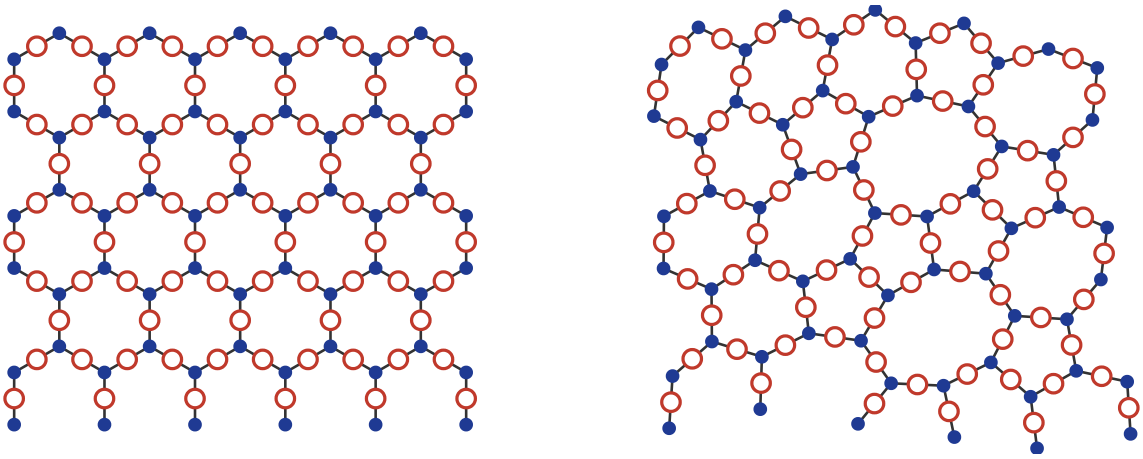


Figure 2. Observation of the Tyndall effect.

The laser beam could not be seen in distilled water, but its path was clearly visible in the gel. The colloidal SiO_2 particles in the gel are close in size to the wavelength of light, so they scatter the beam in all directions. A true solution such as water has no such particles and does not scatter light. This shows that the sol/gel is a colloidal system.

4.3) Crystalline and amorphous SiO_2



(a) Crystalline SiO_2 *(b) Amorphous SiO_2*

● Si ○ O *Figure 3. 2-D schematic of the SiO_4 network.*

Property	Crystalline (quartz)	Amorphous (vitreous)
Order	Long-range, periodic	Only short-range, random rings
Density	2.65 g/cm³	~2.2 g/cm³
Melting	Sharp melting point	Softens gradually (T_g)
XRD / optics	Sharp peaks, anisotropic	Broad hump, isotropic
Chemical	Few Si–OH, less reactive	Many Si–OH, high surface area, more reactive

In both structures each Si is bonded to four O atoms (SiO_4 tetrahedra) and each O bridges two Si atoms; the difference is how the tetrahedra are arranged. **Our powder is amorphous**, because the network formed by random hydrolysis and condensation at room temperature and was heated only to 500 °C, which is far below the crystallization temperature of silica. **Application areas of SiO_2 nanopowder:** filler in polymers and rubber (e.g. tyres), catalyst support, adsorbent and desiccant (silica gel), drug delivery, anti-reflective and water-repellent coatings, polishing slurries, and anti-caking agent in food and cosmetics.

4.4) Contact angle and superhydrophobic surface

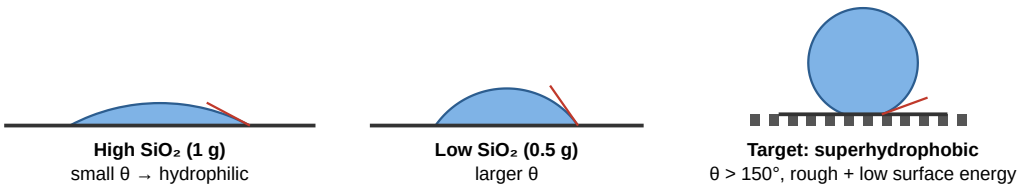


Figure 4. Observed drop shapes (schematic) and the target surface.

A contact angle below 90° means a hydrophilic surface, above 90° hydrophobic and above 150° superhydrophobic. The drop spread more on the high-SiO_2 coating, so **the contact angle decreased and the surface became more hydrophilic as SiO_2 increased**. The reason is the polar Si–OH (silanol) groups on the silica surface, which have high surface energy and form hydrogen bonds with water. The nanofiber paper did not absorb the drop and behaved hydrophobically.

Chemical treatment: to make the surface superhydrophobic, the Si–OH groups must be replaced by low-surface-energy groups. This is done by silanization with a fluoroalkylsilane (FAS) [2]:

$$\equiv Si–OH + (C_2H_5O)_3Si–CH_2CH_2(CF_2)_nCF_3 \rightarrow \equiv Si–O–Si–CH_2CH_2(CF_2)_nCF_3 + C_2H_5OH$$

The fluorinated groups lower the surface energy, and the SiO_2 nanoparticles give roughness that traps air under the drop. Water then touches only the tips of the rough surface and cannot wet it. Brassard et al. treated silica nanoparticles with FAS, spin-coated them and obtained a contact angle of 151° [2]. On such a surface rain drops roll off and carry dirt away (self-cleaning, like the lotus leaf).

4.5) Spin coating: effect of rpm and time

In spin coating, the centrifugal force throws the excess liquid off the substrate while the solvent evaporates. **Higher rpm → thinner film:** the final thickness decreases with spin speed as $h \propto \omega^{-1/2}$ [3]. For example, going from 500 to 800 rpm should make the film about 20 % thinner ($(500/800)^{0.5} \approx 0.79$). **Longer time → thinner only in the first seconds:** most of the thinning happens at the start. After that, solvent evaporation raises the viscosity of the film until it stops flowing, and the thickness no longer changes [3]. This is why 5 s and 10 s gave practically the same coating in our experiment, while the rpm was the controlling parameter. A more viscous solution (more SiO_2) also gives a thicker film.

4.6) Viscosity

Liquid	ρ (g/cm³)	t (s)	η (mPa·s)
Water (ref.)	1.000	1.45	0.89
Low SiO_2	1.02586	8.06	**5.08**
High SiO_2	1.02664	8.34	**5.26**

$$\eta = \eta_0 \cdot (\rho \cdot t) / (\rho_0 \cdot t_0)$$

ρ = m/V: 5.1293/5 = 1.02586 ; 5.1332/5 = 1.02664 g/cm³
Low: 0.89 × (1.02586 × 8.06)/(1 × 1.45) = 5.08 mPa·s
High: 0.89 × (1.02664 × 8.34)/(1 × 1.45) = 5.26 mPa·s

The viscosity increased from 5.08 to 5.26 mPa·s (~3.6 %) as the SiO_2 content increased; both solutions are about 5.7–5.9 times more viscous than water. More solid particles disturb the flow and increase internal friction, and the Si–OH groups on the particles form hydrogen bonds with the –OH groups of PVA. Possible errors are the hand-timed, very short reference time (1.45 s) and uncontrolled temperature.

5) Conclusion

Amorphous SiO_2 nanopowder was synthesized by the sol-gel method with a total weight loss of 96.39 %, and the Tyndall effect showed that the gel is colloidal. Increasing the SiO_2 content made the coating more hydrophilic and increased the viscosity of the PVA solution. In spin coating, a higher speed gives a thinner film, while time matters only in the first seconds. To obtain a superhydrophobic surface, the silica must be silanized with low-surface-energy groups such as FAS.

6) References

1. Experiment 4: The Sol-Gel Technique, MSE4121 Material Production Laboratory Experimental Booklet, Yıldız Technical University, 2026.
2. Brassard, J.-D., Sarkar, D. K., Perron, J., "Synthesis of monodisperse fluorinated silica nanoparticles and their superhydrophobic thin films", *ACS Appl. Mater. Interfaces* 3 (2011) 3583–3588. doi:10.1021/am2007917
3. Meyerhofer, D., "Characteristics of resist films produced by spinning", *J. Appl. Phys.* 49 (1978) 3993–3997. doi:10.1063/1.325357